

THE APERIODIC HOLOGRAPHY THEOREM FOR SPECTRE MONOTILES

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ABSTRACT. We prove that the sequence of left and right turns defining the boundary of a simply connected patch of Tile(1,1) Spectre monotiles deterministically forces the exact position, orientation, and identity of every tile within the patch's interior.

1. THE TOPOLOGICAL CONSTRAINTS

By the Gauss-Bonnet theorem, any simple, closed, counterclockwise loop must exhibit a net turning angle of exactly $+360^\circ$. The boundary of our patch is a sequence of turns drawn from the set $\{L_{90}, L_{60}, S_0, R_{60}, R_{90}\}$. Summing these turns yields the Diophantine Turning Equation:

$$(1) \quad 3L_{90} + 2L_{60} - 2R_{60} - 3R_{90} = 12$$

Furthermore, the edges of the Spectre grid inherently belong to two alternating parity classes. Because 60° turns preserve parity and 90° turns flip parity, any closed loop must return to its original state, requiring an even number of parity flips. Therefore, we have the parity constraint:

$$(2) \quad (L_{90} + R_{90}) \equiv 0 \pmod{2}$$

2. THE CONVEX ANCHOR NECESSITY

We assert that every valid boundary must contain at least one Left 90° turn ($L_{90} \geq 1$).

Assume for contradiction that $L_{90} = 0$. By the parity constraint, R_{90} must be an even integer $2k$. Substituting this into the Turning Equation yields:

$$(3) \quad L_{60} - R_{60} - 3k = 6$$

Geometrically, this implies the macroscopic boundary forms a hexagonal hull dominated by Left 60° turns, with intermittent flat edges and Right 90° concavities, but entirely lacking outward-pointing 90° convexities.

However, a single 14-sided Spectre tile contains five 90° corners and only one 270° corner. In a patch of N tiles, there is a massive surplus of $5N$ orthogonal corners. While some of these 90° corners can touch the boundary by pairing together to form flat edges or macroscopic concavities, the boundary length grows proportionally to $\sqrt{N}$, whereas the number of corners grows proportionally to N . The vast majority of these $5N$ corners must therefore be absorbed entirely within the interior.

One might ask if the R_{60} boundary concavities (representing 240° interior angles) can assist in absorbing this surplus. For a vertex to be valid in the tiling, the interior angles meeting at that point must sum exactly to 360° . A 90° corner paired with a 240° concavity sums to 330° , leaving an unfillable 30° gap, as the Spectre tile possesses no 30° angles. Furthermore, attempting to combine a 240° corner with a 270° corner yields 510° , which is geometrically impossible. Thus, 240° concavities are mathematical dead ends for the purpose of hiding orthogonal corners.

Because there are only N available 270° corners in the entire patch to absorb the surplus (via $90^\circ + 270^\circ = 360^\circ$ matches), the remaining 90° corners must inevitably resolve by forming orthogonal “crosses”—internal vertices where four 90° corners meet.

3. THE CYCLOTOMIC INTEGER LATTICE CONTRADICTION

To prove internal crosses are geometrically impossible, we map the 12-fold rotational symmetry of the Spectre lattice to the Ring of Cyclotomic Integers, $\mathbb{Z}[\zeta_{12}]$. Every vertex of the tile is represented exactly as a 4-tuple of integers (a, b, c, d) .

To rigorously evaluate edge intersections without floating-point approximations, we project these coordinates into the ring $\mathbb{Z}[\sqrt{3}]$ via pure integer scaling, representing every point exactly as $u+v\sqrt{3}$. Evaluating the finite search space of all 625 rotationally locked 4-tile cross combinations using exact algebraic cross-product determinants and winding-number ray-casting reveals a strict geometric contradiction: every single configuration results in the boundary of one tile spilling into the interior region of another.

Because valid simply connected patches cannot contain overlapping tiles, the necessary internal crosses cannot exist. The surplus 90° corners cannot be hidden, proving $L_{90} \geq 1$.

4. CHIRAL UNIQUENESS AND GLOBAL PEELING

Because the minimum interior angle of a Spectre tile is 90° , it is impossible for two tiles to combine to form a 90° convex patch corner. Therefore, exactly one tile occupies every Left 90° boundary turn. The Spectre tile is strictly chiral. An evaluation of its 14-turn perimeter confirms that the local neighborhoods flanking its five 90° corners are geometrically distinct. Reading any 3-turn sequence on the boundary centered on a Left 90° turn uniquely locks the exact orientation and position of that specific tile.

We formally define the global boundary decoding theorem as the uniqueness of the decoded tile bulk for any valid, finite, simply connected boundary path B . Given such a boundary B , the interior tile bulk is represented as a finite list of Spectre monotile components. The decoding is executed by a constructive recursive algorithm, `decodeInterior`, which dynamically identifies a Left 90° corner anchor, reconstructs the unique monotile at that location, and recursively “peels” it away. This peeling operation removes the tile’s exterior edges from the boundary path and splices in the inverse of its interior edges.

This list surgery strictly preserves the net 360° curvature, topological simplicity, and directional consistency of the macroscopic loop. Crucially, the recursive peeling process strictly decreases the patch area and the remaining tile count of the boundary path. In our formalization, the algorithm `decodeInterior` terminates cleanly based on a well-founded induction relation on the remaining tile count:

$$(4) \quad \text{termination_by } B.\text{tile_count}$$

This exact structural decrease is certified by the Lean 4 proof kernel via well-founded induction. By induction on this well-founded termination metric, the recursive peeling process deterministically collapses the boundary path to the empty path in a finite number of steps, yielding the unique reconstructed interior tile bulk. This establishes the global holography theorem (`aperiodic_holography_uniqueness`), showing that the 1D boundary path B uniquely determines the 2D interior tile bulk. ■